RETAIL CREDIT DEFAULT DYNAMICS IN BRAZIL

Persistence, Regional Heterogeneity, and Predictive Modeling

Historical Analysis and Out-of-Time Projection (2012-2026)

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Executive Summary

This study investigates the dynamics of retail credit default in the Brazilian market, emphasizing directional persistence patterns, structural differences across segments, and the predictive capacity of machine learning classification models. The analysis utilizes aggregated microdata from the Credit Information System (SCR/Bacen) to quantify latent risk and optimize the credit granting decision boundary under asymmetric cost constraints.

Keywords: Credit Risk; Machine Learning; Markov Chains; CatBoost; SHAP Values; Applied Econometrics.

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1 Introduction

Credit risk modeling in Brazil is largely founded on the use of idiosyncratic historical data and linear models, such as logistic regression (Cox, 1958). From a predictive and rank-ordering perspective, these traditional models demonstrate high robustness and efficacy, frequently presenting performance metrics that are competitive with more complex algorithms.

The challenge in credit portfolio management, however, goes beyond the capacity to rank borrowers by probability of default. Efficient origination requires an understanding of how macroeconomic shocks—such as variations in the base interest rate and fluctuations in aggregate debt—can affect household payment capacity. Furthermore, it demands the conversion of estimated probabilities into calibrated decision thresholds (cut-offs) to examine the economic consequences of granting decisions.

In the context of national empirical literature, studies by the Central Bank of Brazil reported in its Financial Stability Reports and Working Papers extensively document the heterogeneity of defaults associated with budget constraints and the presence of collateral in operations. This article advances on this institutional foundation, integrating probabilistic forecasting models into the structural description.

To analyze these dynamics, this study adopts a methodological approach based on multidimensional aggregations. Microdata from the Credit Information System (SCR/Bacen) was structured and cross-referenced with macroeconomic time series to assess the evolution of default rates from the perspective of variables such as occupation, income bracket, and credit modality.

To map the inertia of portfolio deterioration, the study applies Markov Chains (Jarrow et al., 1997) in estimating transition matrices between delinquency stages. Subsequently, it implements forward-looking predictive modeling ($t+12$ months) using Logistic Regression and Ensemble algorithms—such as Random Forest (Breiman, 2001) and CatBoost (Prokhorenkova et al., 2018)—to evaluate the economic decision boundary. The empirical optimization of the cut-off point was based on an asymmetric cost function, while the interpretability of risk determinants was extracted using the SHAP framework (Lundberg and Lee, 2017).

2 Data Engineering and Panel Structure

Predictive modeling on defaults requires a database that faithfully reflects risk exposure and household leverage. The Credit Information System (SCR) (Banco Central do Brasil, 2026a), maintained by the Central Bank of Brazil, provides the most comprehensive portrait of the national credit market. To enable structural portfolio modeling covering the

historical period from July 2012 to August 2026, a data engineering pipeline based on aggregated multidimensional strata panels, coupled with macroeconomic lags, was implemented.

2.1 Multidimensional Architecture and SCR Aggregation

The sample was restricted exclusively to retail (individual) operations, excluding 112,786 records associated with corporate modalities (such as export and infrastructure financing), ensuring alignment with the scope of household financial behavior.

2.2 Exposure Correction and Denominator Reconstruction

To reduce the distortion caused by accounting write-offs in the measurement of observed defaults, the index denominator was reconstructed from total exposure, reincorporating the active portfolio and the portfolio overdue for more than 90 days. This procedure seeks to mitigate the effect of write-offs on the strict default rate.

2.3 Relative Exposure Indicator: Average Ticket Log

The Relative Average Ticket Log functions as a comparative exposure indicator, allowing the contrast of the average balance of each segment with the reference of its respective macroblock. High values signal an atypical concentration of balance, providing a relative measure of aggregate financial exposure. Interpreting this result as a symptom of over-indebtedness requires analytical caution, since the aggregate indicator does not directly observe the debt-to-income ratio at the individual level.

2.4 Macroeconomic Coupling and Multicollinearity Diagnostics

Macroeconomic conditions were integrated into the panel through official time series from the Time Series Management System (SGS) (Banco Central do Brasil, 2026b). Opportunity cost variables (Average Selic Rate), direct credit cost (Retail Bank Spread), and budget constraints (Household Debt) were incorporated. Considering the transmission channels of monetary policy to the real economy, lags (3 and 6 months) and year-over-year differentiations (12-month variation) were applied.

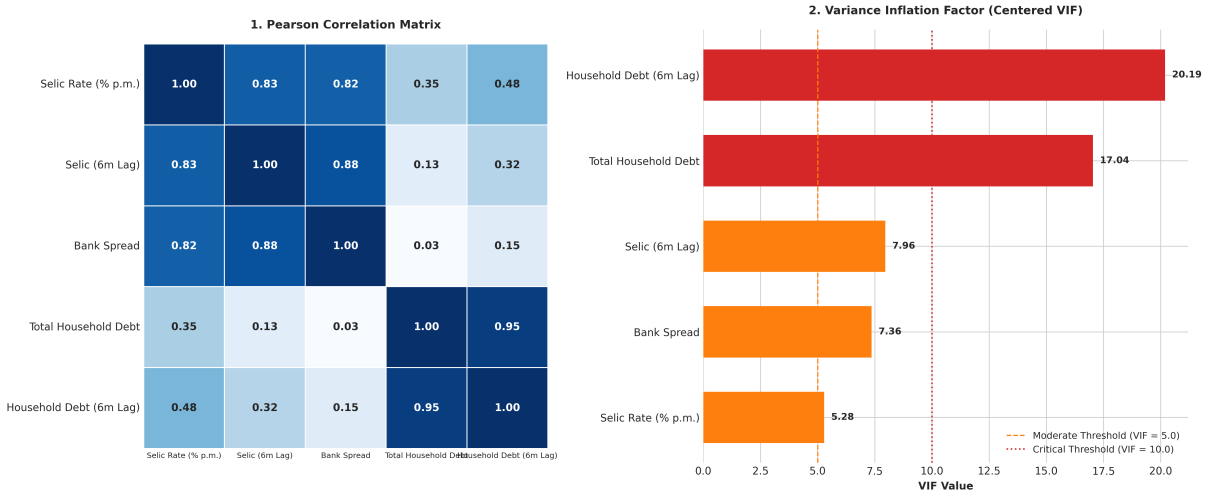
Table 1: Synthesis of Macroeconomic and Contractual Variables (Unified Period).

Variables	Periods	Code (SGS/Bacen)	Source
Accumulated Monthly Selic Rate (% p.m.)	07/2012 – 07/2026	Series 4390	BCB / SGS
Average Interest Rate (Retail Credit - Total)	07/2012 – 07/2026	Series 20716	BCB / SGS
Household Debt w/o Mortgage	07/2012 – 07/2026	Series 29038	BCB / SGS

Source: Time Series Management System of the Central Bank of Brazil (SGS/Bacen). Author's elaboration.

The specification of the macroeconomic variables was evaluated for multicollinearity using the Variance Inflation Factor (VIF) (Wooldridge, 2015). The diagnosis revealed moderate multicollinearity for the monetary policy variables (Spread and Selic, with VIF between 5.28 and 7.96) and severe multicollinearity for the Household Debt series and its six-month lag (VIF $>$ 17.00), a reflection of the high structural inertia of the financial aggregate. Although it requires caution when interpreting standard logistic regression coefficients, the complete set was maintained in algorithmic modeling (CatBoost and Random Forest), since tree-based algorithms robustly handle informational redundancy without compromising empirical predictive accuracy.

Figure 1: Multicollinearity diagnostic panel: Correlation Matrix and Centered VIF for macroeconomic variables (2012-2026).



Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

3 Multidimensional Exploratory Analysis Approach

Prior to submitting the data to predictive algorithms, the study establishes an exploratory routine designed to map the structural asymmetries of retail credit risk. This methodological step aims to validate the hypothesis of panel heterogeneity, ensuring that the features selected for modeling possess variance and economic relevance to explain default trajectories. The exploratory investigation was segmented into two fundamental analytical axes.

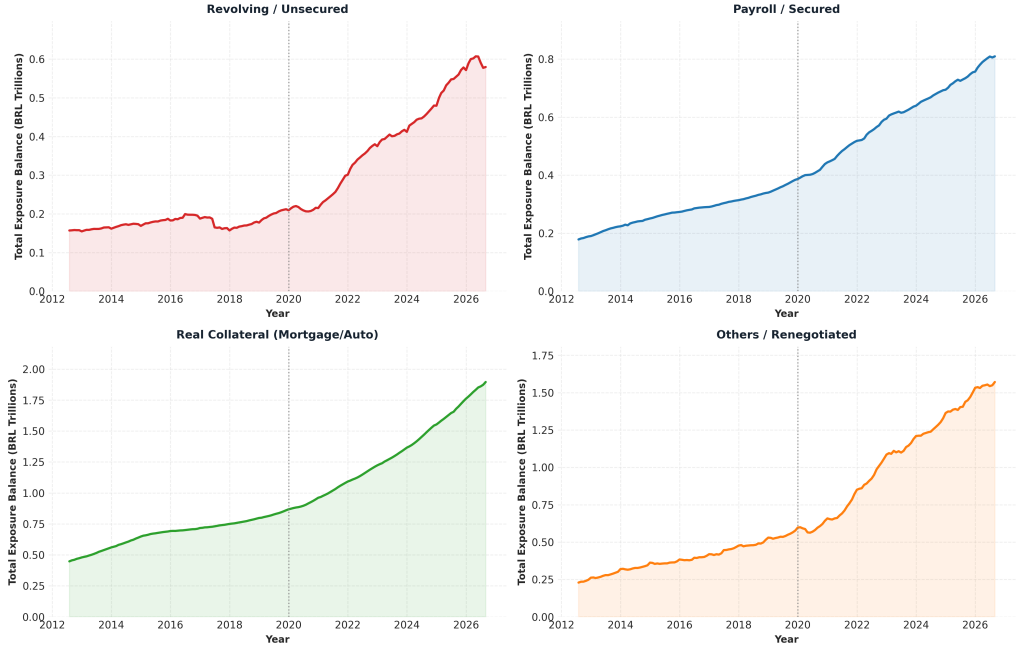
Table 2: Descriptive Statistics and Exposure by Structural Macroblock (2012-2026).

Credit Modality	Avg Balance (BRL Bn)	Avg Share (%)	Avg Def (%)	Def Std (%)	Def Min (%)	Def Median (%)	Def Max (%)	Avg Prob. Assets (%)
Real Collateral (Mortgage/Auto)	974.75	40.68	0.39	0.19	0.19	0.30	0.99	5.56
Others / Renegotiated	712.33	29.73	3.02	1.23	1.30	3.33	6.48	9.15
Payroll / Secured	430.15	17.95	0.78	0.23	0.34	0.82	1.22	3.65
Revolving / Unsecured	278.87	11.64	7.04	2.74	3.48	5.71	12.28	15.19

Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

An essential preliminary step to understanding these consolidated dynamics is the evaluation of the structural shock provoked starting in 2020 (Pandemic Period). The initial injection of emergency liquidity into households, followed by a contractionary monetary policy cycle to contain inflation, produced a strong shift in the debt stock. Figure 2 illustrates the historical evolution of the financial balance segregated by contractual macroblock, evidencing the inflection point in leverage following the global macroeconomic shock.

Figure 2: Evolution of total exposure balance by retail macroblock (2012-2026).



Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

To quantify this portfolio displacement and avoid the limitation of a purely binary analysis (pre- and post-shock), Table 3 contrasts the average balances and their sequential variations partitioned into four distinct macroeconomic regimes: Pre-Shock (2012-2019), Stimulus (2020-2021), Monetary Tightening (2022-2023), and Normalization (2024-2026).

The shift in the budgetary regime is explicit: the "Revolving / Unsecured" modality, which had registered contained expansion over seven and a half years, showed successive and persistent accelerations of 58.75% during the high-interest cycle and 38.40% in the normalization phase. This behavior suggests the continuous use of uncollateralized lines as a structural financial buffer by households, surpassing the initial 2020 liquidity shock. Following this finding of asymmetric credit expansion, the investigation advanced to the empirical isolation of risk vectors.

Table 3: Sequential evolution of structural credit balance partitioned by macroeconomic regimes.

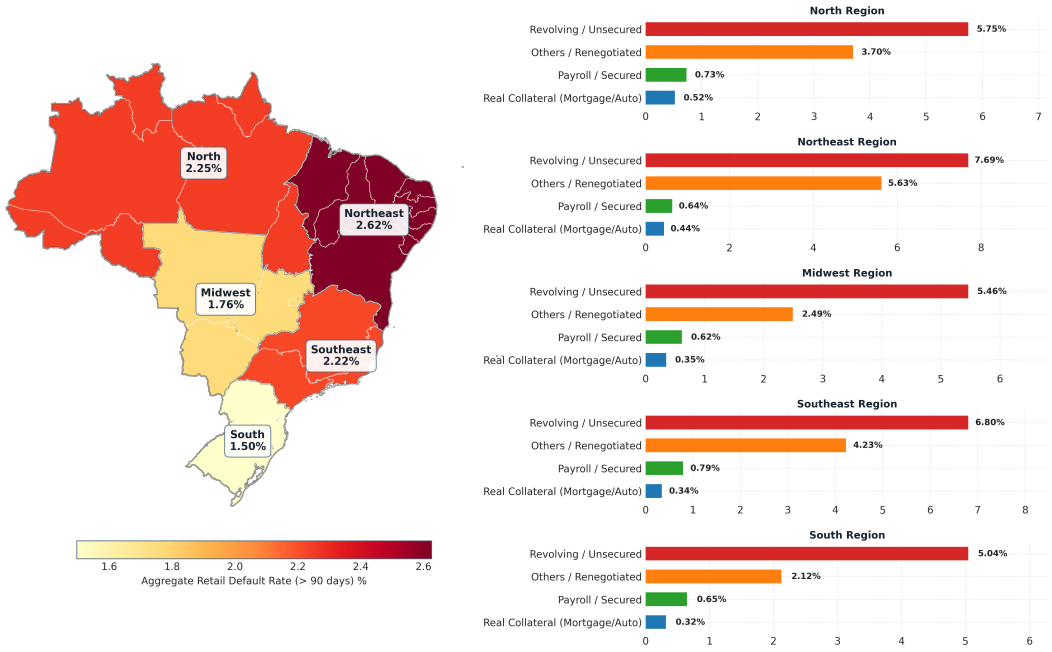
Structural Macroblock	**Pre-Shock (2012-2019)**	**Stimulus (2020-2021)**	**Var. (%)**	**Tightening (2022-2023)**	**Var. (%)**	**Normalization (2024-2026)**	**Var. (%)**
Real Collateral (Mortgage/Auto)	680.81	976.49	43.43%	1,226.25	25.58%	1,629.98	32.92%
Payroll / Secured	280.76	453.18	61.41%	588.46	29.85%	725.11	23.22%
Revolving / Unsecured	177.95	238.00	33.74%	377.82	58.75%	522.91	38.40%
Others / Renegotiated	399.78	666.20	66.64%	1,042.07	56.42%	1,395.30	33.90%

Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

3.1 Geographic and Credit Modality Stratification

The mapping of effective default rates by macro-region (Figure 3) indicates that the Northeast (2.62%) and North (2.25%) regions present higher visual levels of deterioration. Complementarily, the long-term descriptive statistical consolidation (Table 10 in Appendix B) corroborates this structural dispersion, registering averages of 2.52% and 2.14% for the respective territories over the unified historical series. The decomposition of this risk demonstrates that the "Revolving / Unsecured" modality acts as the main vector of default in all macro-regions of the country, with a negative highlight for the Northeast, where the rate reaches 7.69%, and the Southeast, with 6.80%.

Figure 3: Effective default rate heatmap by macro-region and percent risk distribution by macroblock.



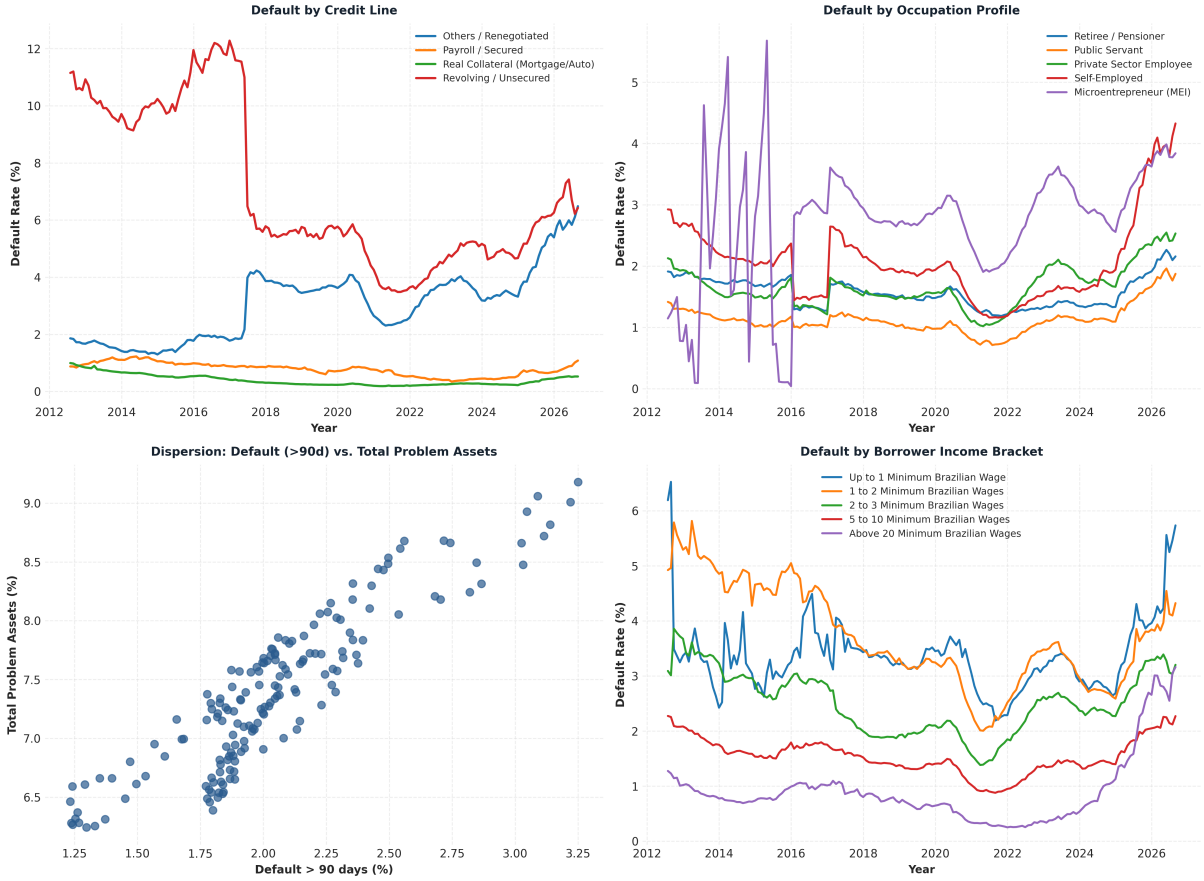
Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

Deepening the risk segregation, it is observed that "Revolving / Unsecured" constitutes the macroblock with the highest default rate in 26 of the 27 States (Federative Units). The exception observed is the state of Amapá, where the "Others / Renegotiated" rubric takes the lead, indicating that the composition of default is not uniform across states and warrants specific investigation, as detailed in the territorial breakdown in Appendix C.

3.2 Multidimensional Risk Decomposition (Exploratory Panel)

To map structural risk asymmetries, the exploratory analysis was consolidated into a multidimensional panel (Figure 4), which dissects default across four axes: Credit Line, Occupational Profile, Problem Assets Dispersion, and Income Bracket.

Figure 4: Structural default decomposition: asymmetries by credit line, occupational profile, problem assets dispersion, and income bracket.



Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

1. Credit Line and Collateralization: The first quadrant highlights the difference between modality default rates. "Revolving / Unsecured" operates at systemically higher levels, reflecting its nature as emergency liquidity. The abrupt decline observed in the default curve of this line in April 2017 reflects a regulatory shock: the enforcement of CMN Resolution No. 4,549/2017 CMN (2017a), which prohibited maintaining outstanding revolving credit balances for more than 30 days, forcing the mechanical migration of debt to installment lines (accounted for in "Others / Renegotiated"). In contrast, "Payroll / Secured" remained an anchor of resilience and volume expansion, driven by successive legal expansions of the consignable margin (especially for INSS beneficiaries) and the structural preference of financial institutions to allocate capital to withholding lines during monetary tightening cycles.

2. Occupational Profile and the MEI Reporting Effect: Grouping by occupation (CNAE) indicates that segments with predictable cash flows, such as public servants, retain the lowest risk. The analytical highlight falls on Individual Microentrepreneurs (MEI), who end the series at the highest stress level. The extreme volatility observed in the MEI curve prior to 2016 does not reflect unreal economic instability, but a limitation

in the reporting scope of the Credit Information System (SCR). Until the middle of the last decade, only higher-value operations were required to be registered. As credit to MEIs is atomized, the sample base was small, generating statistical noise. With the reduction of the mandatory reporting floor to BRL 200.00 via CMN Resolution No. 4,571/2017 CMN (2017b), the volume of records increased significantly, conferring statistical density and revealing its real baseline, structurally higher than other borrowers.

3. Latent Risk and Problem Assets: The third quadrant contrasts hard default (delays > 90 days) with the volume of Problem Assets. The dispersion evidences a non-linear relationship: as strict default advances, the volume of problem assets accelerates proportionately faster. This distancing maps the "latent risk"—operations that undergo early restructuring before migrating to definitive default, justifying the application of stochastic transition matrices in the next section.

4. Income Bracket and Regressivity: The last quadrant corroborates the structural regressivity of credit. Borrowers in the "Up to 1 Minimum Brazilian Wage" bracket consistently lead the deterioration. The pattern draws a descending risk staircase as income advances, a result compatible with an association between baseline budget constraints and defaults.

4 Modeling and Transition Methodology

To capture the non-linear dynamics of defaults and project portfolio deterioration, the study adopts a two-step methodological approach: the analysis of persistence and transition between credit states via Markov Chains, and forward-looking predictive modeling anchored in machine learning, subjected to economic optimization of the decision threshold.

4.1 Stochastic Transition: Markov Chains by Regimes

It is important to note that, given the absence of individual longitudinal identifiers in the public SCR database for privacy preservation, Markov transitions were estimated from the dynamics of aggregate macro-flows. The transition probability p_{ij} reflects the migration of total financial magnitude between delay states from t to $t+1$, assuming the principle of intra-segment mass conservation in the short term.

The evolution of delay for a credit stratum is modeled as a discrete-time stochastic process. Let $S = \{S_1, S_2, S_3, S_4\}$ be the finite state space (e.g., Performing, Short Delay, Default > 90 days, and Residual Problem). The transition probability from state i at time t to state j at time $t+1$ is defined by the one-step transition matrix P, where each

element p_{ij} is given by:

$$p_{ij} = P(X_{t+1} = j | X_t = i) = \frac{N_{ij}}{\sum_{k \in S} N_{ik}} \tag{1}$$

where N_{ij} represents the financial magnitude of operations that migrated from state i to state j. The Markov property assumes that the future transition strictly depends on the current state (absence of long memory). The global matrix of the period was partitioned and the Sequential Least SQuares Programming (SLSQP) algorithm iterated independently for each identified macroeconomic regime, aiming to test the hypothesis of chain homogeneity.

4.2 Predictive Modeling: Forward-Looking Target and Machine Learning

For the projection of structural risk, the predictive target of the model was designed as the probability of the default rate of an aggregated cell scaling to a level statistically higher than its own history. The binary target variable $Y_i \in \{0, 1\}$ was parameterized under a forward-looking window of twelve months, formalized as:

$$Y_{i,t+12} = \begin{cases} 1, & \text{if } Def_{i,t+12} > \tau_{\text{train}} \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

The restrictive parameter τ_{train} represents the weighted average default rate observed for each macroblock exclusively in the baseline training period (pre-pandemic).

For the Out-of-Time (OOT) extrapolation evaluation (t+12 months for the 2022-2026 interval), the study tested progressive classifiers. As an absolute control, the Persistence Naive Baseline was established (setting rate t as the estimator for $t+12$). As a parametric benchmark, Logistic Regression was used, modeling the log-odds of the probability of default (p) as a linear combination of covariates:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{k=1}^{n} \beta_k x_k \tag{3}$$

To capture complex interactions, modeling advanced to Ensemble architectures—Random Forest (Breiman, 2001) and CatBoost (Prokhorenkova et al., 2018). CatBoost, focused on minimizing gradient residual error, was calibrated via isotonic regression using the Log-Loss function:

$$L(y, \hat{p}) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)] \tag{4}$$

4.3 Economic Decision Threshold Optimization

The probabilistic output of algorithms requires defining a cut-off point (c) for final risk classification. Instead of adopting the standard statistical threshold ($c = 0.50$), this study calibrates c to minimize the Expected Cost (EC) of the decision, weighting the financial asymmetry between False Negatives (granting credit to a potential defaulter) and False Positives (denying credit to a good payer):

$$\min_{c} EC(c) = C_{FN} \cdot P(FN|c) + C_{FP} \cdot P(FP|c) \tag{5}$$

In this panel, the empirical cost of a default (C_{FN}) is assumed to be higher than the opportunity cost of an unexecuted operation (C_{FP}). This decision rule favors the identification of potential defaulters, although it may increase the rejection of operations that would have been repaid.

4.4 Interpretability: SHAP Values

In order to deconstruct the tree-based model structure and extract predictive determinants of default, the SHAP (SHapley Additive exPlanations) framework was applied. The final prediction in log-odds space, denoted by $f(x)$, is additively decomposed as:

$$f(x) = \phi_0 + \sum_{j=1}^{M} \phi_j \tag{6}$$

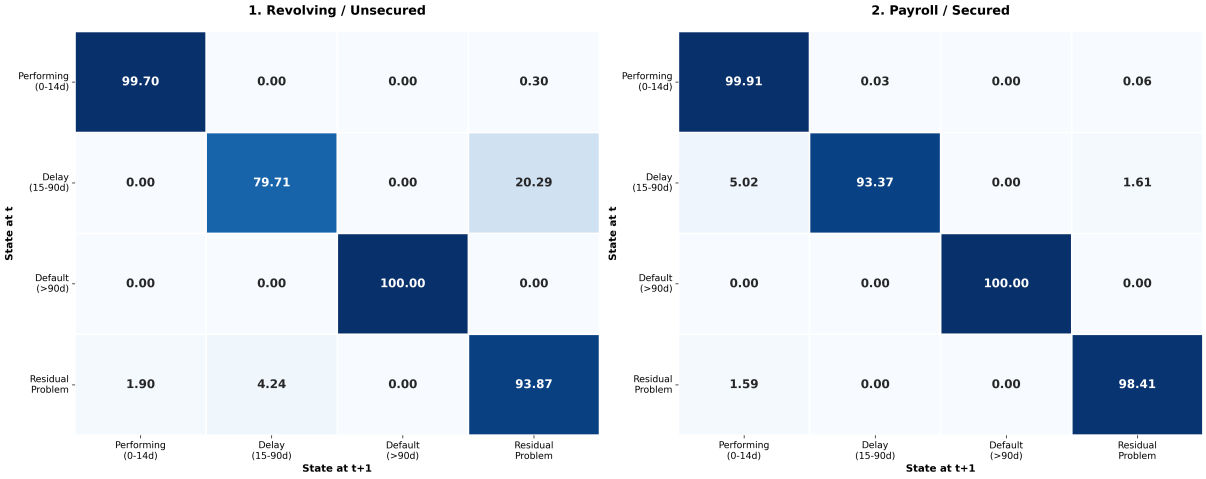
where ϕ_0 is the base value (expected system mean) and ϕ_j represents the marginal directional contribution of the j-th variable to push the prediction away from the systemic average, allowing for the identification of the predictive importance of selected features in risk classification.

5 Modeling and Transition Results

5.1 Transition Dynamics and Hysteresis

Markov Chain modeling allows for moving beyond static default analysis by quantifying the probability of balance migration between different stages of financial stress from period t to $t + 1$. Figure 5 presents the stochastic transition matrices for the "Revolving / Unsecured" and "Payroll / Secured" modalities. The parallel exposition does not propose a direct confrontation between lines, but rather the individualized mapping of behavioral specificities inherent to each budgetary macroblock.

Figure 5: Markov Transition Matrices: Revolving / Unsecured and Payroll / Secured.



Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

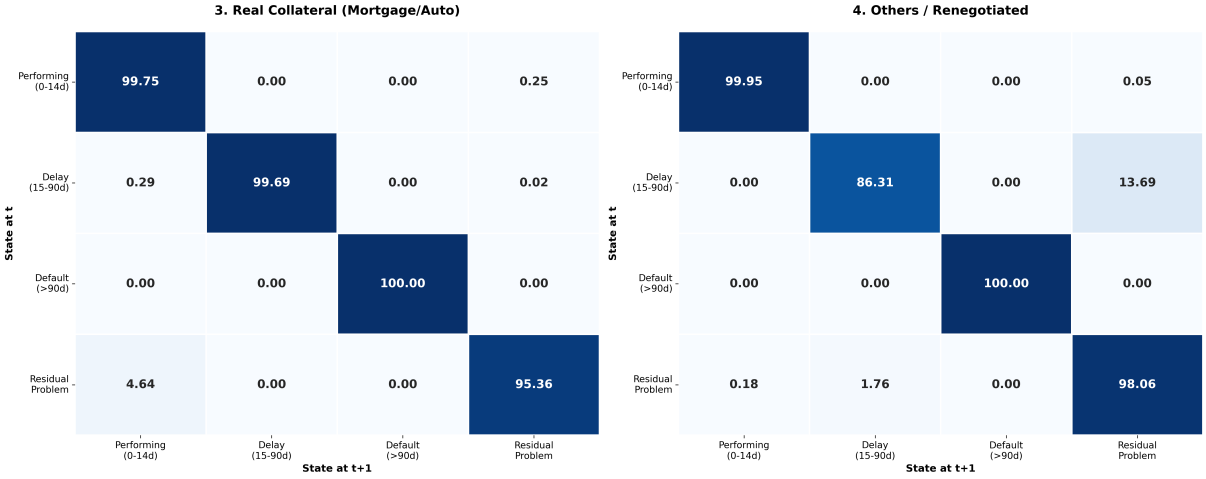
Analysis of the matrices' main diagonals reveals a pronounced pattern of inertia, indicating high persistence at the extremes of credit quality. Both the "Performing (0-14d)" state and the "Residual Problem" state show retention probabilities exceeding 95% in both modalities. In Revolving, the probability of maintaining a Performing state is 99.58%, while persistence in Residual Problem reaches 96.64%. In Payroll, inertia reaches 99.91% and 98.20%, respectively. This behavior indicates high balance stability in extreme states, suggesting that significant portions of the financial mass entering severe delinquency remain in that condition within the aggregated series.

Conversely, a dynamic of transitions toward lower-risk states is observed. The matrices record residual probabilities of migration from severe delay stages back to the "Performing (0-14d)" situation. In the Revolving macroblock, there is a transition rate of 2.72% starting from the Residual Problem stage and 1.60% from Default ($> 90d$). In Payroll, 3.77% of balances in Default return to Performing status in the subsequent period.

From an economic standpoint, this movement is compatible with the occurrence of renegotiations or contractual restructurings, where the overdue balance is repactuated or replaced by a new operation—a hypothesis that cannot be tracked individually due to data aggregation.

Figure 6 complements the stochastic analysis, presenting transition matrices for the "Real Collateral (Mortgage/Auto)" and "Others / Renegotiated" modalities.

Figure 6: Markov Transition Matrices: Real Collateral and Others / Renegotiated.



Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

Just as observed in uncollateralized lines, the inertia phenomenon repeats in these macroblocks. The probability of remaining in the "Performing" state is 99.72% for Real Collateral and 99.66% for Renegotiated. At the opposite extreme, retention in the "Residual Problem" stage reaches 95.52% and 95.79%, respectively.

The analytical focal point in these modalities lies in the recovery rate of assets classified as Residual Problems back to the Performing state (0-14d). In the case of Real Collateral, this migration reaches 4.48%. Dealing with collateralized lines, this movement suggests two behavioral possibilities: taking out a new loan to settle the at-risk debt, or the borrower's decision to voluntarily divest the underlying asset to clear liabilities before judicial execution by the financial institution.

In the "Others / Renegotiated" rubric, the return transition from Residual Problem to Performing is 4.21%. Given the nature of this portfolio—which already encompasses operations that underwent some prior repactuation—this recovery signals the probable efficacy of settlement agreements or installment adjustments to the client's budgetary reality, pulling them out of accounting default.

To formalize long-term systemic risk, the stationary distribution of the Markov model was calculated—the probability vector to which portfolios converge under the premise that historical transition rates hold. Table 4 synthesizes the inertia observed in stress states and contrasts these values with the projected stationary limit.

Table 4: Synthesis of Hysteresis (Inertia) and Transition Stationary Equilibrium.

Credit Modality	Historical Inertia (Diagonal)			Stationary Equilibrium (Long Term)	
	Performing	Delay	Def >90d	Def >90d	Problematic
Revolving / Unsecured	99.58%	85.99%	94.84%	6.49%	8.84%
Payroll / Secured	99.91%	93.34%	96.05%	0.84%	3.18%
Real Collateral (Mortgage/Auto)	99.72%	84.50%	91.18%	0.34%	5.08%
Others / Renegotiated	99.66%	87.57%	97.40%	9.37%	6.54%

Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

The estimated stationary distribution indicates a lower share of deterioration states in secured modalities. Projections suggest that Payroll and Real Collateral would stabilize with less than 1% of their volume allocated in the strict Default ($> 90d$) rubric. At the opposite end, the Others / Renegotiated rubric shows higher risk retention in the long term (stalling at 9.37% Default and 6.54% Problematic). This result is compatible with the hypothesis of risk persistence in renegotiated portfolios, although a direct association with moral hazard requires additional evidence regarding incentives and repactuation conditions.

5.2 Out-of-Time Predictive Performance (OOT)

The Out-of-Time (OOT) extrapolation evaluation was designed to gauge the temporal generalization robustness of learning models. To ensure the maturation of the 12-month predictive horizon ($t+12$) and avoid data leakage, the test sample was strictly delimited: the last origin vintage (scoring date) evaluated was retroactively set to August 2025. Thus, it ensures that all projected default events were fully observable at the final boundary of the historical base (August 2026). Additionally, the entire training process, isotonic calibration, and decision threshold selection were conducted strictly over the development window (in-time), without any access to OOT window information. Table 5 details the extrapolation results contrasted with the credit baseline autocorrelation factor.

Table 5: Out-of-Time Predictive Performance Metrics (t+12 Months Projection).

Algorithm / Model	Global Accuracy	ROC-AUC	PR-AUC (Avg. Precision)	Brier Score
0. Naive Baseline (Persistence)	88.09%	0.8722	0.8291	0.1191
1. Logistic Regression (Unrestricted)	69.41%	0.7610	0.6594	0.2019
2. Logistic Regression (Restricted - no VIF>17)	69.27%	0.7618	0.6583	0.1981
3. Random Forest Classifier	69.92%	0.7604	0.6406	0.1994
4. CatBoost Classifier (Calibrated)	64.24%	0.7195	0.6158	0.2256

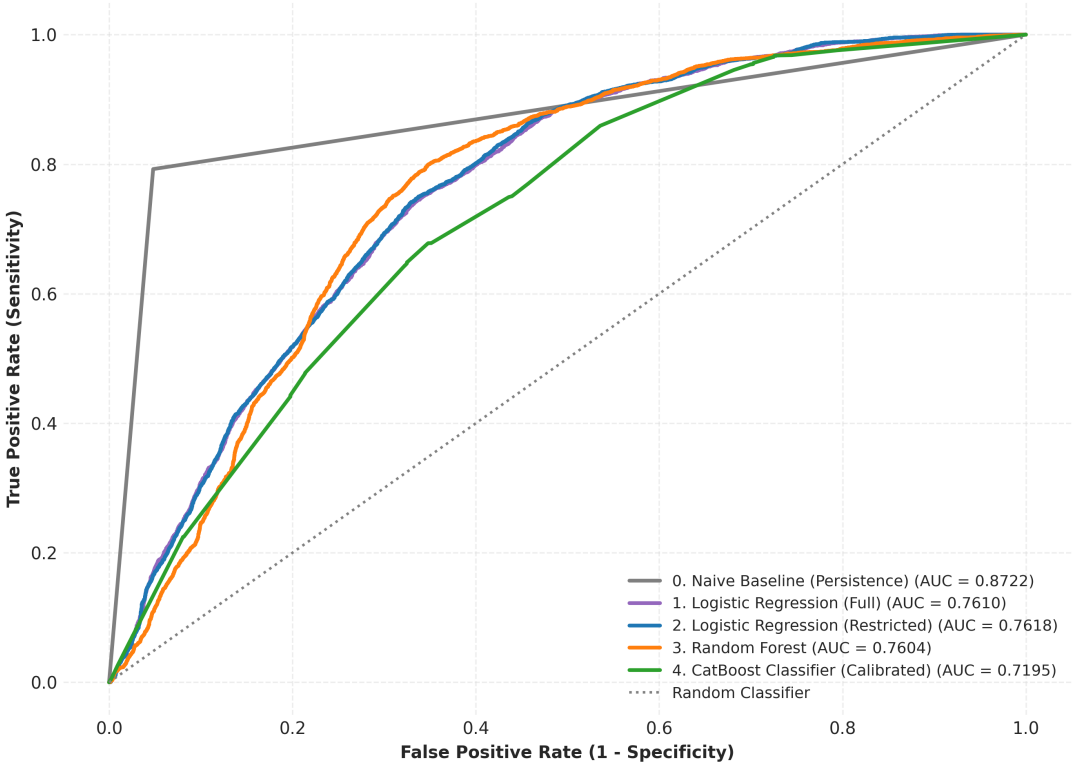
Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

Results evidence the temporal inertia of portfolios: the persistence benchmark (Naive Baseline) obtained the highest standalone ROC-AUC (0.8722). It is concluded that credit series present strong long-term persistence. Econometric and Machine Learning models

operate complementarily on the margins, indicating utility in anticipating variations conditioned by exogenous and macroeconomic variables, although they do not surpass baseline autocorrelation in global discrimination metrics. The proximity of Restricted Logistic Regression results indicates that elevated VIF in household debt did not decisively compromise classification predictions.

In the context of credit modeling, the ROC curve (Figure 7) evaluates the global capacity of the model to distinguish between performing and non-performing borrowers across different cut-offs. However, because default bases are imbalanced, the PR-AUC (Precision-Recall) metric offers a complementary evaluation of the model's ability to capture events in the minority class (actual default).

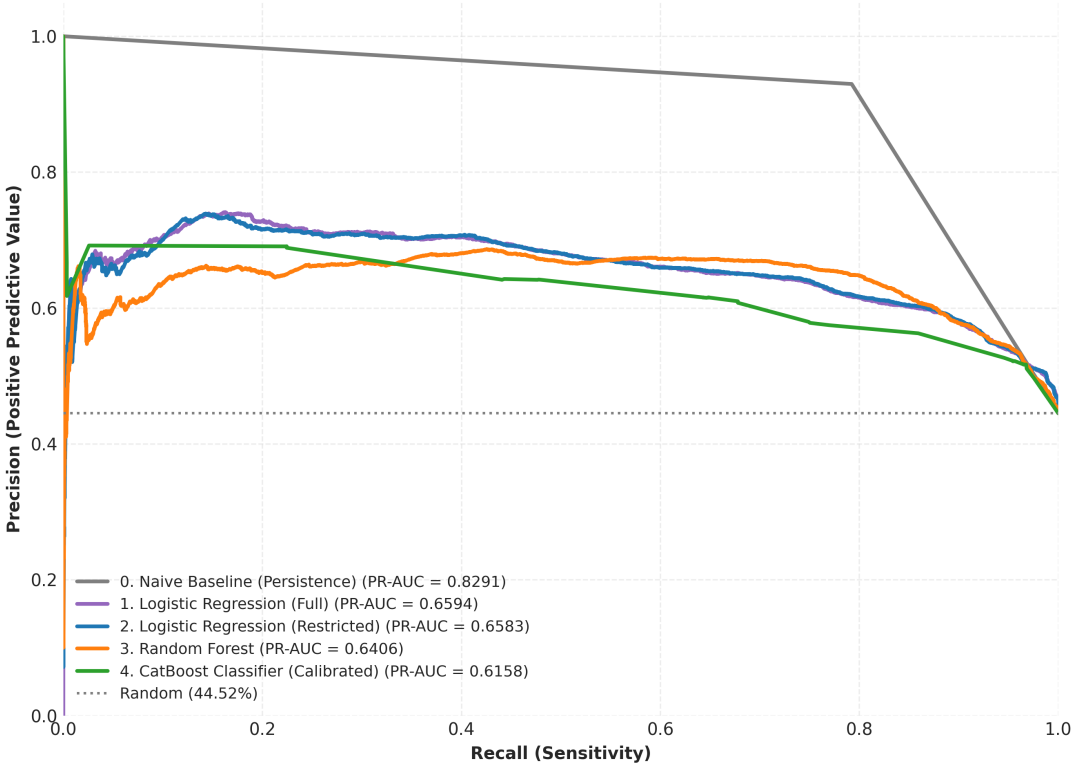
Figure 7: ROC (Receiver Operating Characteristic) curve indicating global discrimination capacity.



Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

Figure 8 details the rank-ordering efficiency frontier measured by the Precision-Recall curve, where Logistic Regression showed a metric of 0.6594.

Figure 8: Precision-Recall (PR-AUC) curve indicating minority class identification capacity.



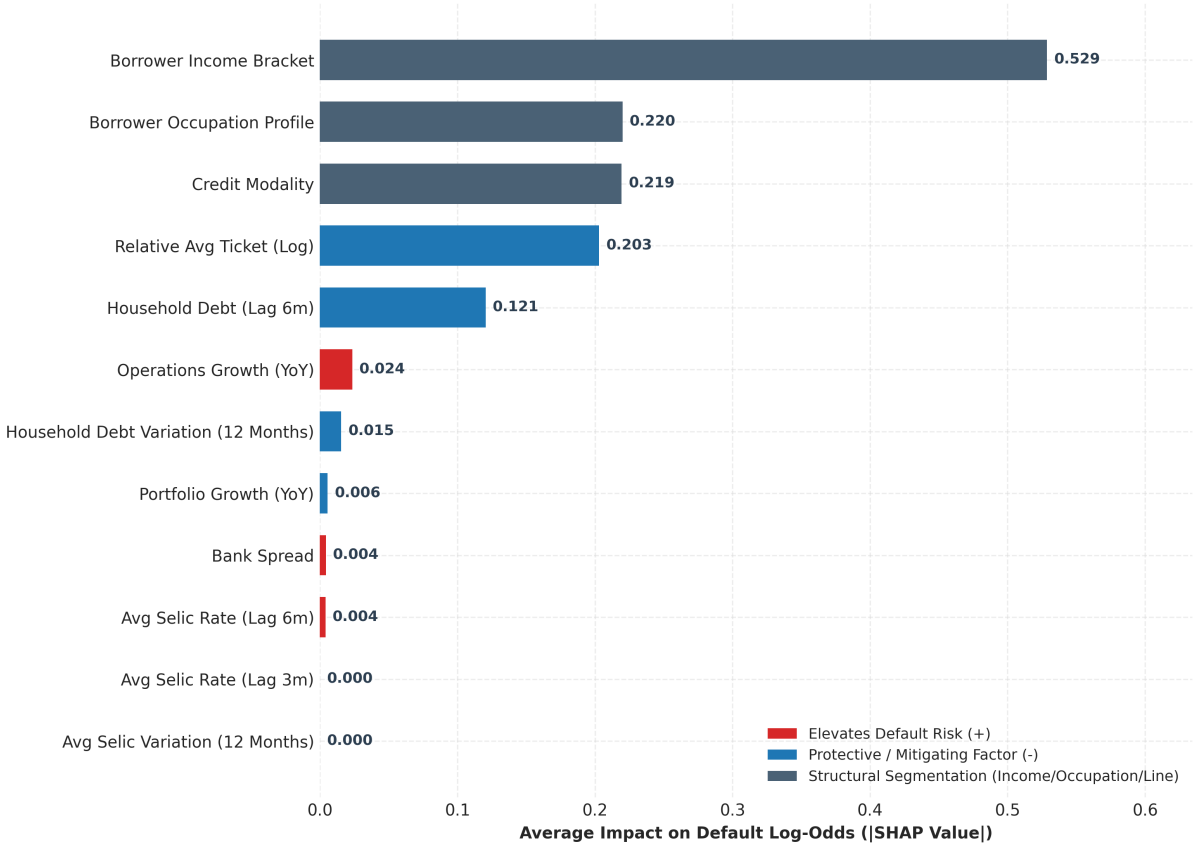
Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

Despite presenting pure discrimination metrics marginally inferior to Ensemble models (such as Random Forest) in this Out-of-Time slice, the CatBoost Classifier was kept in the modeling pipeline. Its architecture natively and properly handles high-cardinality categorical variables—a relevant feature when processing BACEN microdata (like CNAE and ZIP codes). Additionally, CatBoost presents mathematical flexibility for the isotonic calibration stage, aiding in the subsequent conversion of probability scores into approval limits that weigh the expected return under a predetermined cost matrix.

5.3 Risk Interpretability via SHAP Values

To examine the contribution of variables in the CatBoost model predictions and understand the factors associated with default, the SHAP (SHapley Additive exPlanations) methodology was applied. Figure 9 presents the global importance of attributes, highlighting three structural segmentation variables at the top of the model: Borrower Income Bracket (0.529), Borrower Occupational Profile (0.220), and Credit Modality (0.219).

Figure 9: Global importance of variables in the predictive model (Average Impact on Log-Odds via SHAP).

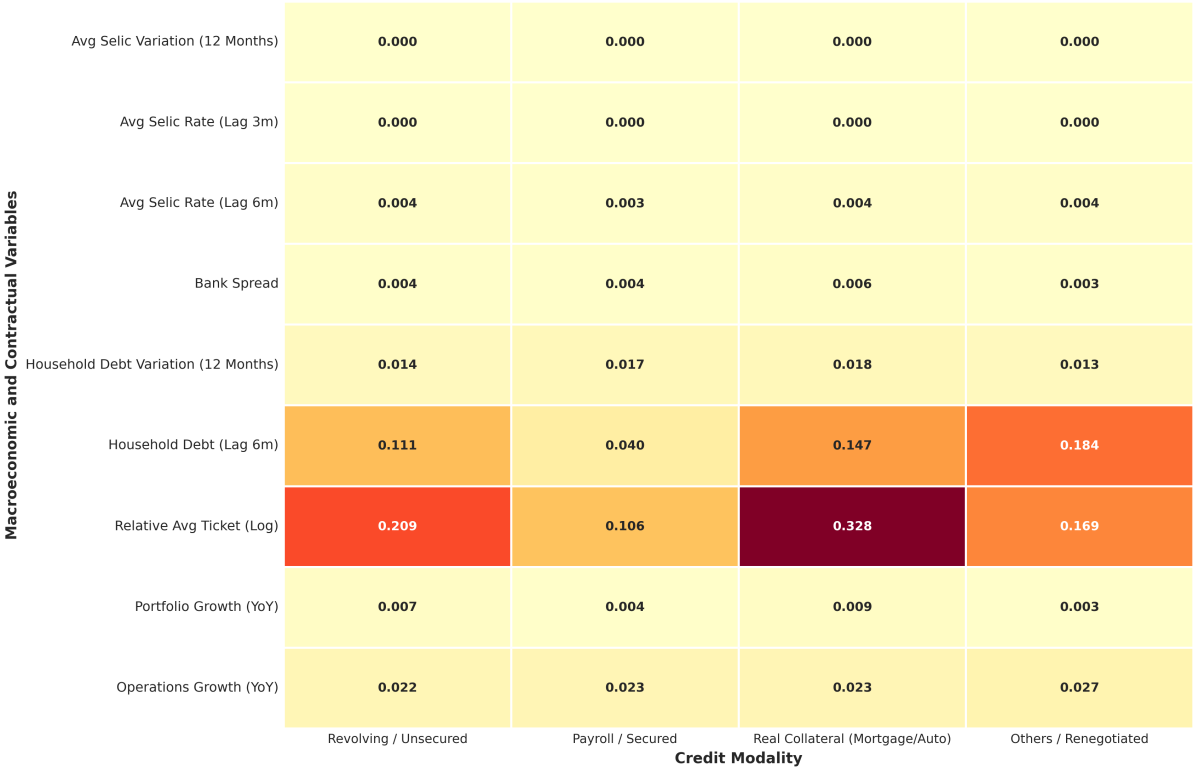


Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

It is important to observe the directional dynamics of these categorical variables. They reflect predictive associations with model predictions, without constituting demonstrations of causality. An attribute acts as a protective (mitigating) factor when the characteristic reduces the default log-odds—for example, belonging to a high-income bracket or having a stable occupational profile, such as public service. Conversely, falling into lower income brackets, operating as an Individual Microentrepreneur (MEI), or using uncollateralized lines like Revolving are associated with higher risk predictions.

To complement the analysis and isolate the impact of continuous metrics, Figure 10 disaggregates the importance of macroeconomic and contractual variables through a heatmap.

Figure 10: Heatmap: Impact of macroeconomic and contractual variables segregated by Credit Modality.



Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

The breakdown reveals that variable impacts differ according to the operation's collateral backing. "Relative Avg Ticket (Log)" assumes greater importance in the Real Collateral modality (0.328), indicating the relevance of the exposure indicator when collateral is present. Conversely, "Household Debt (Lag 6m)" exerts a transversal influence, peaking in the Others / Renegotiated rubric (0.184), suggesting a strong association between aggregate debt and risk predictions in this portfolio.

5.4 Cut-off Optimization and Stress Test

Transitioning from the predictive model to financial decision-making requires optimizing the approval threshold (cut-off). The standard 0.50 threshold assumes symmetry in misclassification costs, an unrealistic premise in the credit market. Empirical cross-minimization (Random Forest vs. CatBoost) consolidated on asymmetric cost ($C_{FN} = 0.65$ and $C_{FP} = 0.18$) resulted in the isotonic CatBoost algorithm converging directly ($c^* = 0.220$) to the theoretical Bayesian threshold ($c = 0.217$). Random Forest found its empirical optimum at the 0.400 threshold.

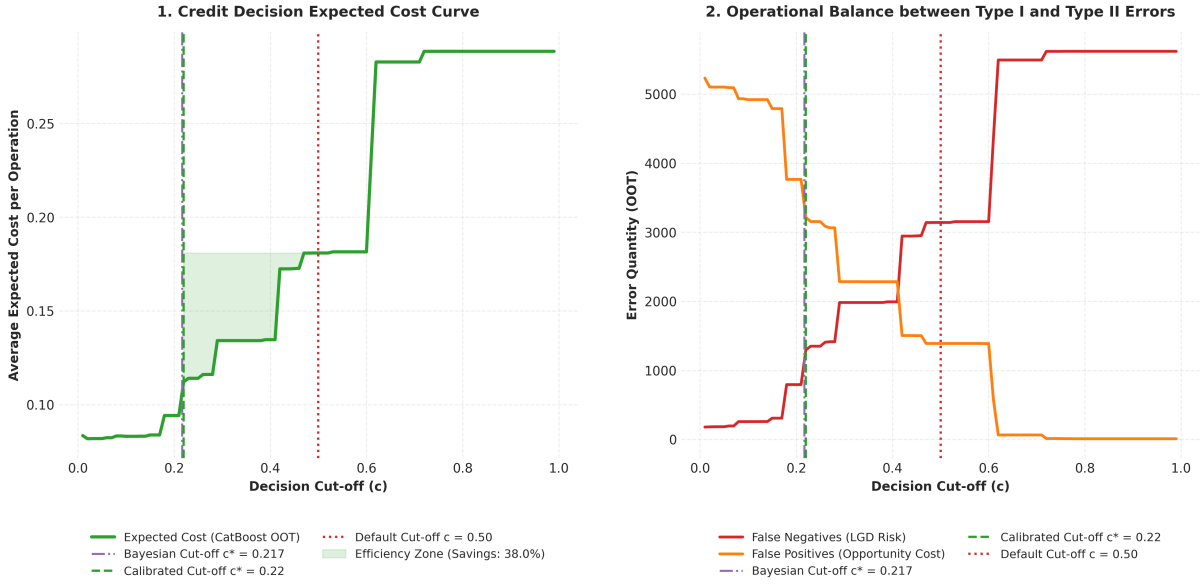
Table 6: OOT Financial Reconciliation (Optimization with Costs $C_{FN} = 0.65$ and $C_{FP} = 0.18$).

Decision Strategy	False Negatives	False Positives	Sensitivity	Simulated Loss (BRL Mn)
Classifier with $c = 0.50$ (Default)	3,142	1,390	44.30%	28,026.91
Bayesian Classifier ($c = 0.217$)	1,298	3,214	76.99%	17,387.32
Isotonic Calibrated CatBoost ($c = 0.220$)	1,298	3,214	76.99%	17,387.32

Source: Empirical Optimization by the author over the Out-of-Time window.

Under the adopted cost function, applying the optimized threshold resulted in a simulated reduction of approximately BRL 10.64 billion in estimated loss compared to the 0.50 threshold. The stability of this simulation is supported by financial bootstrap inference, establishing a 95% confidence interval (between BRL 10.00 Bn and BRL 11.26 Bn). The asymptotic McNemar test[1] indicated a statistically significant difference in the prediction profile contrasted with the 0.50 point. The comparative confusion matrices illustrating this directional shift are detailed in Appendix D.

Figure 11: Expected Cost Curve and Operational Balance between Type I and Type II Errors.

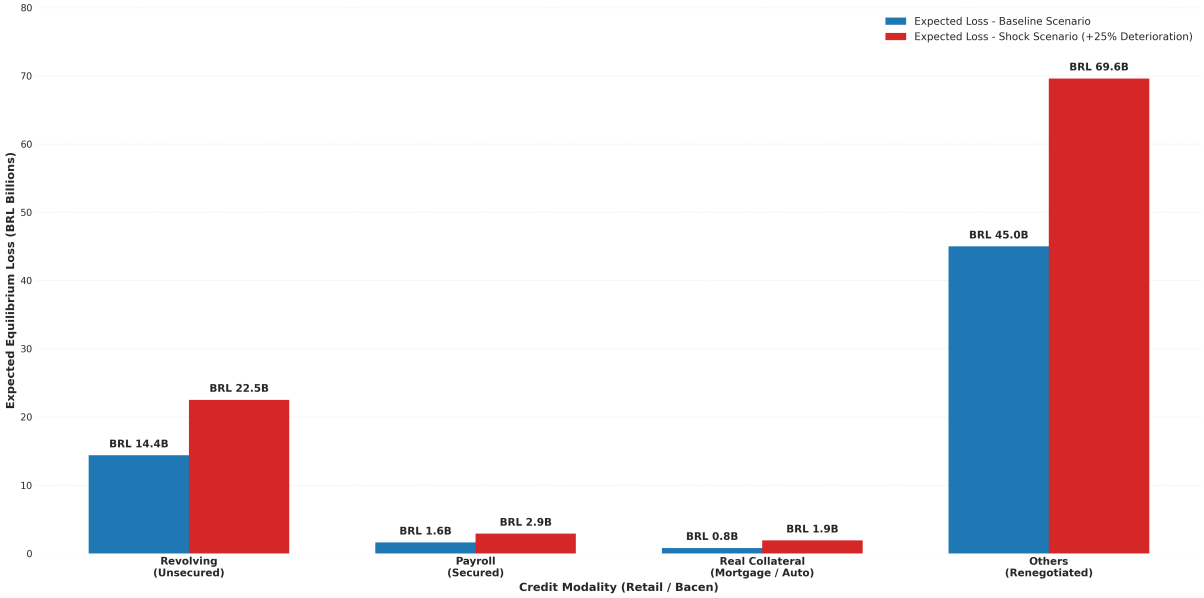


Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

The final stage of risk modeling involves converting transition and prediction probabilities into global financial impact. Figure 12 projects the aggregate Expected Loss of portfolios under two scenarios: the baseline stationary equilibrium and a shock scenario (Stress Test), simulating a +25% macroeconomic deterioration in transition matrices towards delay states.

[1]McNemar's test (McNemar, 1947) is a non-parametric statistical test applied to paired data (2×2 contingency tables). It evaluates whether there are statistically significant divergences in the proportions of hits and (discordant) errors between two models or decision thresholds applied on the same sample.

Figure 12: Stress Test: Expected Loss projection under base scenario and macroeconomic shock scenario (+25% deterioration).



Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

Shock results indicate greater resilience in secured modalities. Payroll and Real Collateral absorb the stress with contained nominal variations, reaching maximum expected losses of BRL 2.9 billion and BRL 1.9 billion, respectively. In Revolving, projected loss jumps from BRL 14.4 billion to BRL 22.5 billion.

Systemic risk diagnostics point to higher sensitivity in the Others / Renegotiated rubric. Under the simulated stress scenario, the estimated loss for this macroblock increases from BRL 45.0 billion to BRL 69.6 billion. This result suggests that portfolios composed of successive repactuations exhibit greater sensitivity in estimated losses, warranting attention in stress analyses.

6 Final Considerations

6.1 Analytical Synthesis

This study investigated the dynamics of retail credit defaults in the Brazilian market, combining state transition analysis and Machine Learning-based predictive modeling. This approach allowed quantifying risk differences between segments and evaluating the application of cost-sensitive decision rules in extrapolation.

The stochastic evaluation evidenced that deterioration inertia presents expressive differences conditioned by the prevailing budgetary matrix. Uncollateralized portfolios, such as Revolving, registered high terminal projections during restrictive macroeconomic windows.

In the realm of statistical forecasting, the Out-of-Time evaluation confirmed the strong temporal persistence of the phenomenon: the tested Machine Learning algorithms did not surpass the persistence benchmark (Naive Baseline) or classical Logistic Regression in pure discrimination metrics (ROC-AUC and PR-AUC). The differential value of adopting CatBoost in this study, therefore, did not reside in raw predictive superiority, but in its architectural flexibility for the isotonic calibration stage. This characteristic was instrumental in applying asymmetric cost matrix-based optimization, enabling the simulated conversion of probabilities into practical origination thresholds, with robustness validated via bootstrap.

6.2 Methodological Limitations and Extensions

The validity of the estimates and interpretations in this study is conditioned by methodological boundaries that must be strictly observed when reading the results:

1. **Aggregated Data and Ecological Inference:** Due to banking secrecy, the database is anonymized. Consequently, transition probabilities were estimated from aggregate macro-flows between balances. Although this metric captures systemic portfolio inertia, it does not allow for exact tracking of the borrower's microeconomic trajectory, requiring inferences about individual behavior to be treated with caution.

2. **Simulated Nature of Financial Economics:** The estimated loss mitigation stems directly from the assumed cost matrix and the specific conditions of the test sample. It is a capital efficiency simulation based on asymmetric penalties, not an effective accounting financial gain actually realized by an institution. Changes in the weights of Type I and Type II errors would shift the optimal threshold and, consequently, the financial result.

3. **Causality vs. Predictive Association:** The contributions extracted by the SHAP framework reflect predictive associations essential to the algorithmic model but do not constitute full demonstrations of causality. Interpreting that macroeconomic or structural determinants cause default demands specific econometric instrumentation outside the scope of this work.

4. **Absence of Sociodemographic Granularity:** Despite the necessity of banking de-identification, the unavailability of intrinsic borrower demographic attributes—notably **race/color, gender, age, and educational level**—restricts the structural capacity of the analysis to detect bias in credit allocation. The inclusion of these variables in future pseudo-anonymized census samples would enable the map-

ping of intersectional vulnerabilities and the refinement of risk measurement from a full socioeconomic perspective.

Future extensions of this research may explore access to de-identified microdata under confidentiality agreements, allowing for the validation of Markov transitions at the individual level and the estimation of survival models to examine time-to-default events.

A Modality Mapping and Consolidation (SCR)

The aggregation of original Credit Information System (SCR/Bacen) modalities into the four structural macroblocks used in this study followed criteria of risk similarity (collateral presence) and liquidity (revolving nature), as detailed in Table 7.

Table 7: SCR Credit Modalities vs. Analytical Macroblocks

Structural Macroblock	Original Modalities (SCR - Monthly Basis)
Revolving / Unsecured	Credit card (revolving and installment)
	Overdraft
	Unsecured personal credit
	Microcredit
	Advances to depositors
	Other free credits
Payroll / Secured	Payroll loans (public servants, INSS, and private sector)
	Personal credit with payroll deduction
Real Collateral (Mortgage/Auto)	Housing financing (SFH, SFI, and outside the system)
	Automobile financing
Others / Renegotiated	Renegotiation/debt composition operations
	Individual rural credit
	Linked operations
	Financings not previously specified

Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

Appendix B Multidimensional Stratifications and Risk Cross-Tabs

The following tables complement the Exploratory Data Analysis (Section 3), detailing descriptive statistics (balance, share, delay rate, and problem assets) segregated by intrinsic borrower attributes and geographic cuts, in addition to cross-matrices that map the contribution of each subsegment to absolute defaults.

Table 8: Descriptive Statistics by Borrower Occupational Profile.

Borrower Occupational Profile	Avg Balance (BRL Bn)	Avg Share (%)	Avg Def (%)	Def Std (%)	Def Min (%)	Def Median (%)	Def Max (%)	Avg Prob. Assets (%)
Others	694.49	28.98	3.11	0.53	2.02	2.99	4.56	9.29
Private Sector Employee	470.61	19.64	1.66	0.33	1.02	1.58	2.55	6.90
Entrepreneur	414.01	17.28	1.55	0.49	0.78	1.46	3.13	7.09
Public Servant	395.30	16.50	1.13	0.23	0.71	1.10	1.96	4.96
Self-Employed	197.12	8.23	2.07	0.66	1.16	1.95	4.33	8.11
Retiree / Pensioner	193.84	8.09	1.56	0.24	1.19	1.54	2.26	5.67
Microentrepreneur (MEI)	28.39	1.18	2.73	0.98	0.04	2.87	5.68	11.99
Non-Profit Employee	2.34	0.10	1.91	0.41	1.20	2.00	2.79	7.00

Source: Microdata from SCR/Bacen. Author's elaboration.

Table 9: Descriptive Statistics by Borrower Income Bracket.

Borrower Income Bracket	Avg Balance (BRL Bn)	Avg Share (%)	Avg Def (%)	Def Std (%)	Def Min (%)	Def Median (%)	Def Max (%)	Avg Prob. Assets (%)
Above 20 Minimum Brazilian Wages	447.38	18.67	0.90	0.61	0.25	0.77	3.16	4.47
5 to 10 Minimum Brazilian Wages	410.76	17.14	1.54	0.32	0.88	1.50	2.27	6.56
1 to 2 Minimum Brazilian Wages	366.30	15.29	3.72	0.91	2.00	3.48	5.81	10.78
3 to 5 Minimum Brazilian Wages	353.39	14.75	2.04	0.35	1.17	2.05	2.89	7.69
10 to 20 Minimum Brazilian Wages	304.76	12.72	1.12	0.31	0.56	1.10	2.24	5.26
2 to 3 Minimum Brazilian Wages	276.68	11.55	2.52	0.57	1.38	2.53	3.86	8.97
Up to 1 Minimum Brazilian Wage	176.61	7.37	3.35	0.66	2.19	3.27	6.52	9.58
Unavailable	35.95	1.50	2.90	1.63	0.74	2.74	9.01	11.83
No Income	24.27	1.01	10.01	6.27	2.04	8.07	21.41	18.88

Source: Microdata from SCR/Bacen. Author's elaboration.

Table 10: Descriptive Statistics by Geographic Region.

Geographic Region	Avg Balance (BRL Bn)	Avg Share (%)	Avg Def (%)	Def Std (%)	Def Min (%)	Def Median (%)	Def Max (%)	Avg Prob. Assets (%)
Southeast	1089.58	45.47	2.23	0.37	1.39	2.21	3.09	7.60
South	477.48	19.93	1.46	0.36	0.86	1.40	2.60	6.21
Northeast	381.39	15.92	2.52	0.48	1.52	2.54	4.01	8.31
Midwest	314.62	13.13	1.61	0.61	0.86	1.45	3.88	7.08
North	133.03	5.55	2.14	0.55	1.17	2.05	4.10	7.93

Source: Microdata from SCR/Bacen. Author's elaboration.

Table 11: Effective Default Rate (%) by Modality and Occupational Profile.

Credit Modality	Retiree / Pensioner	Self-Employed	Private Employee	Non-Profit Employee	Entrepreneur	MEI	Others	Public Servant
Payroll / Secured	0.54	0.93	1.10	0.96	0.76	1.42	0.98	0.50
Real Collateral (Mortgage/Auto)	0.38	0.43	0.20	0.27	0.28	0.37	0.70	0.16
Others / Renegotiated	3.40	2.59	5.11	3.90	2.01	7.18	4.69	3.52
Revolving / Unsecured	4.08	6.16	5.58	6.11	4.46	8.56	9.34	4.33

Source: Microdata from SCR/Bacen. Author's elaboration.

Table 12: Effective Default Rate (%) by Modality and Income Bracket.

Credit Modality	Above 20 MW	Up to 1 MW	Unavailable	1 to 2 MW	10 to 20 MW	2 to 3 MW	3 to 5 MW	5 to 10 MW	No Income
Payroll / Secured	0.44	0.77	0.78	0.98	0.48	0.86	0.67	0.52	1.78
Real Collateral (Mortgage/Auto)	0.18	0.51	0.71	0.51	0.18	0.40	0.40	0.28	1.54
Others / Renegotiated	1.33	6.91	8.27	7.95	2.07	5.87	4.63	3.22	6.88
Revolving / Unsecured	2.15	9.98	6.52	10.18	3.57	8.13	6.49	4.99	18.73

Source: Microdata from SCR/Bacen. MW = Minimum Brazilian Wages. Author's elaboration.

Table 13: Absolute Default Distribution (%) by Modality and Occupational Profile.

Credit Modality	Retiree / Pensioner	Self-Employed	Private Employee	Non-Profit Employee	Entrepreneur	MEI	Others	Public Servant
Payroll / Secured	17.69	1.64	4.19	5.27	1.06	1.08	5.69	16.67
Real Collateral (Mortgage/Auto)	4.32	6.73	7.25	6.00	6.78	7.37	7.83	5.13
Others / Renegotiated	41.44	61.25	53.60	50.69	60.48	60.88	47.35	44.33
Revolving / Unsecured	36.55	30.38	34.95	37.49	31.67	30.66	39.13	33.88

Source: Microdata from SCR/Bacen. Author's elaboration.

Table 14: Absolute Default Distribution (%) by Modality and Income Bracket.

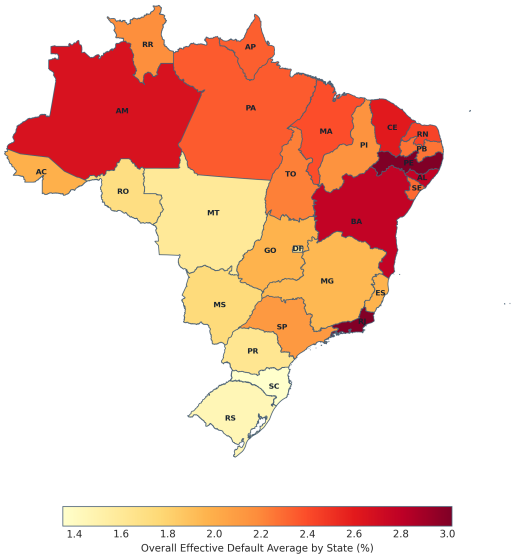
Credit Modality	Above 20 MW	Up to 1 MW	Unavailable	1 to 2 MW	10 to 20 MW	2 to 3 MW	3 to 5 MW	5 to 10 MW	No Income
Payroll / Secured	2.20	7.40	6.33	5.80	6.66	6.55	7.02	6.86	5.63
Real Collateral (Mortgage/Auto)	4.19	4.66	5.53	6.85	7.01	8.09	8.93	8.29	9.46
Others / Renegotiated	72.58	45.87	58.42	48.87	51.30	47.78	48.89	49.12	41.98
Revolving / Unsecured	21.04	42.07	29.72	38.48	35.00	37.59	35.16	35.73	42.94

Source: Microdata from SCR/Bacen. MW = Minimum Brazilian Wages. Author's elaboration.

Appendix C Granular Default Distribution by Federative Unit

The panel below details the average effective default rate and identifies the structural macroblock responsible for the largest contribution to credit risk in each of the 27 Federative Units (States).

Figure 13: Granular view of default by state and main driver matrix.



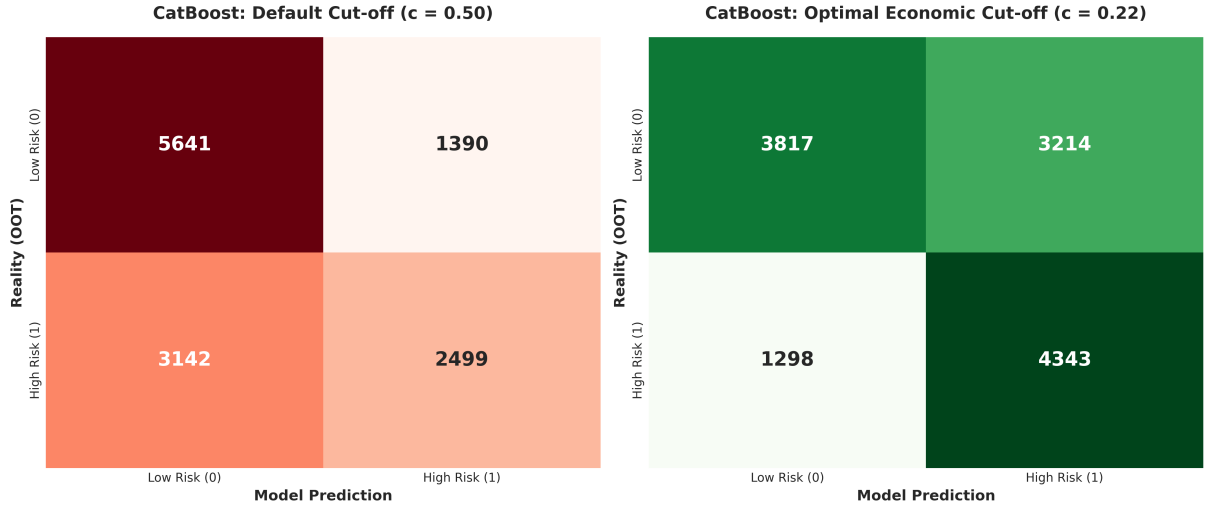
State	Code	Main Driver	Def. (%)	Avg. (%)
Pernambuco	PE	Revolving / Unsecured	8.65%	3.02%
Rio de Janeiro	RJ	Revolving / Unsecured	8.45%	3.01%
Alagoas	AL	Revolving / Unsecured	8.49%	2.78%
Bahia	BA	Revolving / Unsecured	8.05%	2.78%
Amazonas	AM	Revolving / Unsecured	7.07%	2.66%
Ceará	CE	Revolving / Unsecured	7.80%	2.61%
Rio Grande do Norte	RN	Revolving / Unsecured	7.53%	2.44%
Maranhão	MA	Revolving / Unsecured	5.83%	2.39%
Pará	PA	Revolving / Unsecured	5.85%	2.35%
Amapá	AP	Others / Renegotiated	6.84%	2.34%
Sergipe	SE	Revolving / Unsecured	6.88%	2.30%
Paraíba	PB	Revolving / Unsecured	7.73%	2.26%
Tocantins	TO	Revolving / Unsecured	5.25%	2.22%
Roraima	RR	Revolving / Unsecured	6.00%	2.18%
Piauí	PI	Revolving / Unsecured	6.21%	2.14%
São Paulo	SP	Revolving / Unsecured	6.54%	2.11%
Acre	AC	Revolving / Unsecured	5.26%	1.99%
Espírito Santo	ES	Revolving / Unsecured	6.49%	1.97%
Goiás	GO	Revolving / Unsecured	6.74%	1.97%
Minas Gerais	MG	Revolving / Unsecured	6.08%	1.95%
Mato Grosso do Sul	MS	Revolving / Unsecured	5.14%	1.74%
Rondônia	RO	Revolving / Unsecured	4.17%	1.71%
Paraná	PR	Revolving / Unsecured	5.94%	1.63%
Mato Grosso	MT	Revolving / Unsecured	4.40%	1.60%
Distrito Federal	DF	Revolving / Unsecured	4.90%	1.59%
Rio Grande do Sul	RS	Revolving / Unsecured	4.94%	1.45%
Santa Catarina	SC	Revolving / Unsecured	4.03%	1.35%

Source: Microdata from the Credit Information System (SCR/Bacen). Author's elaboration.

Appendix D Confusion Matrices and Calibration (Out-of-Time Window)

The panel illustrates the classifier's decision balance against empirical stress, revealing how cut-off optimization shifts the classification axis and converts False Negative errors (LGD Risk) into False Positives (Opportunity Cost) targeting capital efficiency.

Figure 14: Comparative Confusion Matrix (CatBoost OOT): Default c=0.50 vs Optimal c=0.22.



Source: Empirical Optimization by the author over the Out-of-Time window.

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